

Calculus I Lesson 01

1) $f(x) = \frac{1}{x-2}$

a) Complete the following table.

x	$f(x)$
1	
1.5	
1.9	
1.999	
1.999999	
1.999999999	

b) Complete the following table.

x	$f(x)$
3	
2.5	
2.1	
2.0001	
2.0000001	
2.0000000001	

Using TI-84: Graphing Function:
 Press [Y=]
 Input: $Y1 = 1/(X-2)$
 Press [GRAPH]

Finding Function Values:
 Press [2nd] [WINDOW]
 Set Indpnt on Ask



Press [2nd] [GRAPH]
 Input x-values: 1, 1.5, 1.9, 1.999, 1.999999, 1.999999999

c) $\lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{1cm}} ?$

limit of $f(x)$ as x approaches 2 from the left

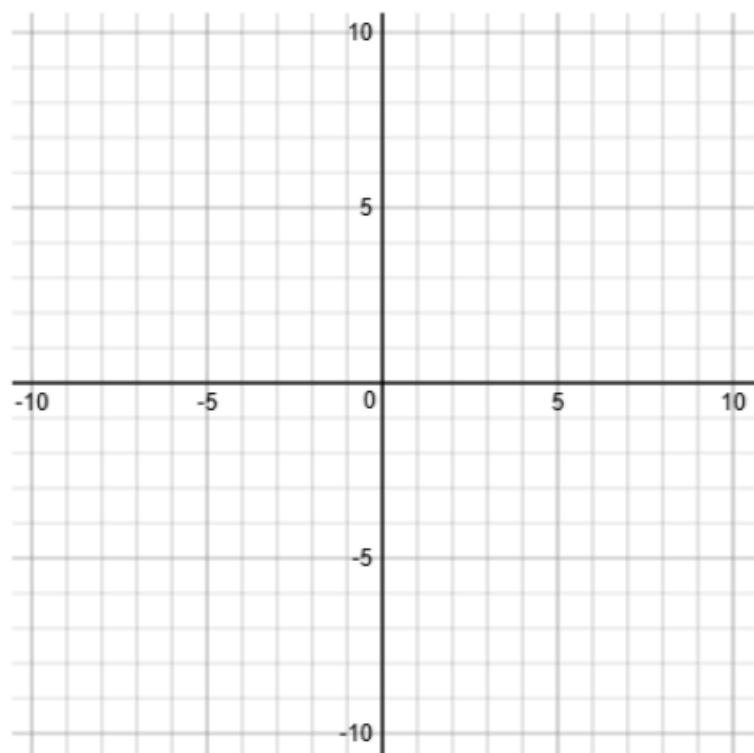
$\lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{1cm}} ?$

limit of $f(x)$ as x approaches 2 from the right

$\lim_{x \rightarrow 2} f(x) = \underline{\hspace{1cm}} ?$

limit of $f(x)$ as x approaches 2

d) Graph $f(x) = \frac{1}{x-2}$



$$2) f(x) = \frac{x}{x-1}$$

a) Complete the following table.

x	$f(x)$
0	
0.5	
0.9	
0.999	
0.999999	
0.999999999	

b) Complete the following table.

x	$f(x)$
2	
1.5	
1.1	
1.0001	
1.0000001	
1.0000000001	

c) $\lim_{x \rightarrow 1^-} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches 1 from the left

$\lim_{x \rightarrow 1^+} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches 1 from the right

$\lim_{x \rightarrow 1} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches 1

c) $\lim_{x \rightarrow 1^-} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 1 from the left

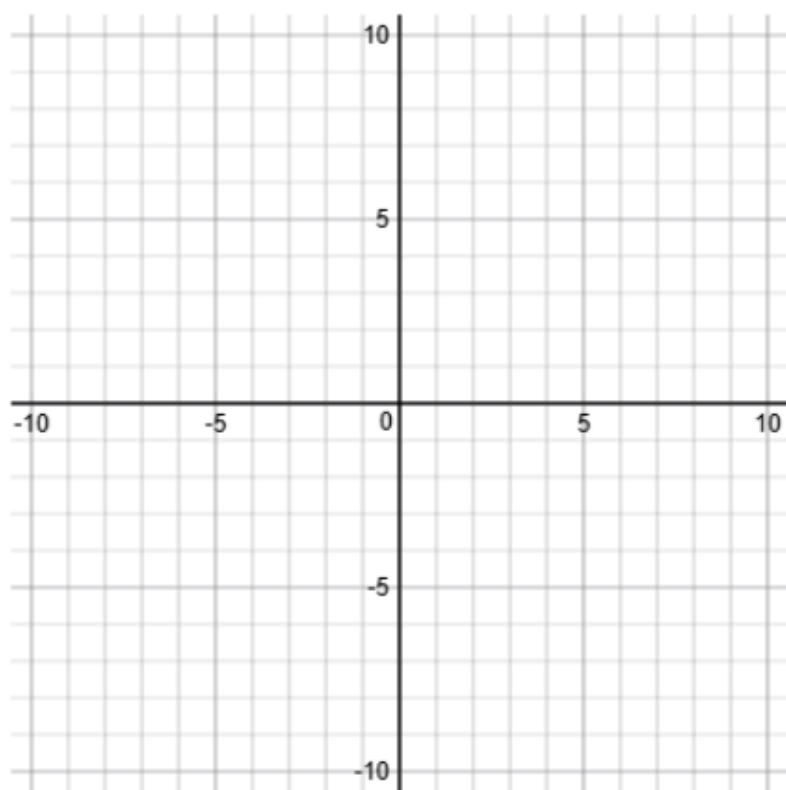
$\lim_{x \rightarrow 1^+} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 1 from the right

$\lim_{x \rightarrow 1} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 1

d) Graph $f(x) = \frac{x}{x-1}$



$$3) f(x) = \frac{3x}{2x+4}$$

a) Complete the following table.

x	$f(x)$
-3	
-2.5	
-2.1	
-2.0001	
-2.0000001	
-2.0000000001	

b) Complete the following table.

x	$f(x)$
-1	
-1.5	
-1.9	
-1.999	
-1.999999	
-1.999999999	

c) $\lim_{x \rightarrow -2^-} f(x) = \underline{\hspace{1cm}} ? \hspace{1cm}$

limit of $f(x)$ as x approaches -2 from the left

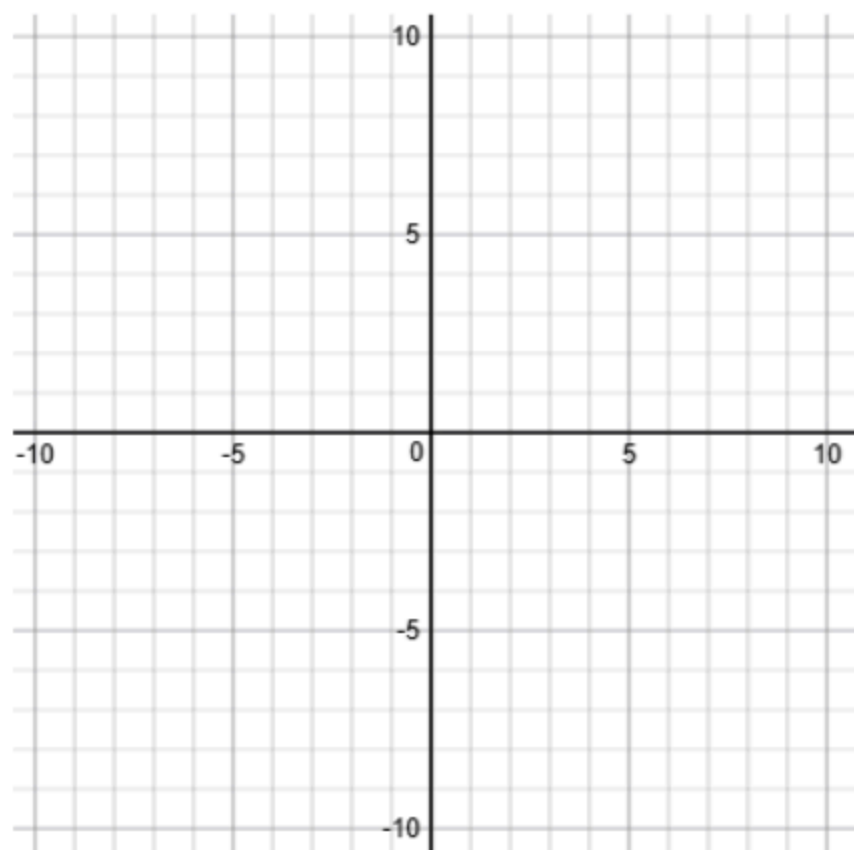
$\lim_{x \rightarrow -2^+} f(x) = \underline{\hspace{1cm}} ? \hspace{1cm}$

limit of $f(x)$ as x approaches -2 from the right

$\lim_{x \rightarrow -2} f(x) = \underline{\hspace{1cm}} ? \hspace{1cm}$

limit of $f(x)$ as x approaches -2

d) Graph $f(x) = \frac{3x}{2x+4}$



$$4) f(x) = \frac{3x}{2x-6}$$

a) Complete the following table.

x	$f(x)$
2	
2.5	
2.9	
2.999	
2.999999	
2.999999999	

b) Complete the following table.

x	$f(x)$
4	
3.5	
3.1	
3.0001	
3.0000001	
3.0000000001	

c) $\lim_{x \rightarrow 3^-} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches 3 from the left

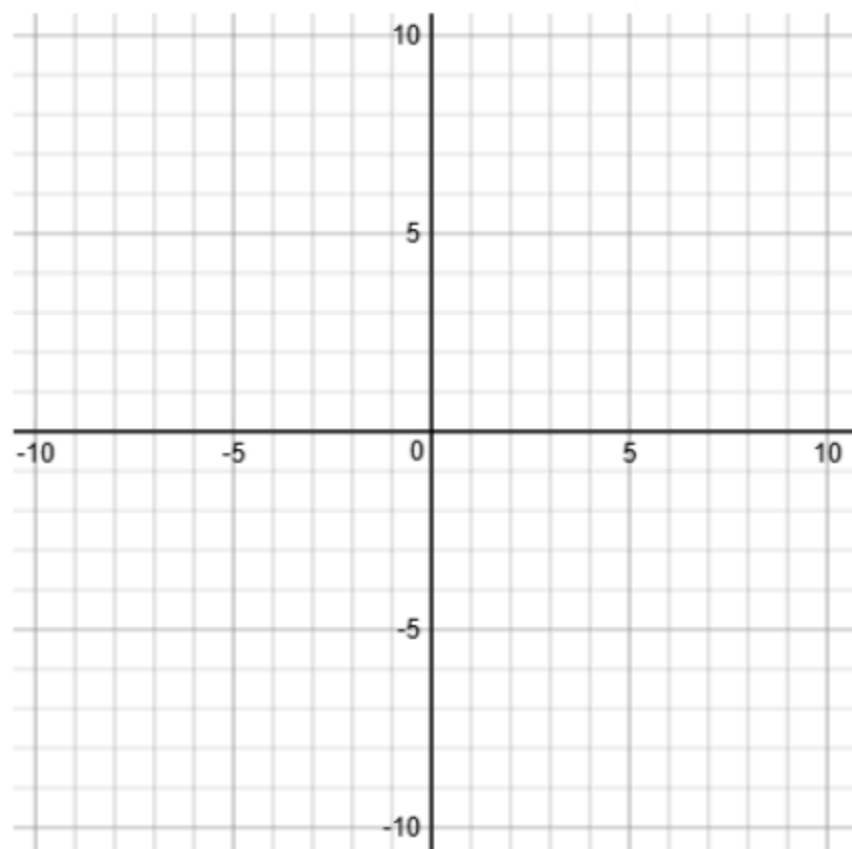
$\lim_{x \rightarrow 3^+} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches 3 from the right

$\lim_{x \rightarrow 3} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches 3

d) Graph, $f(x) = \frac{3x}{2x-6}$

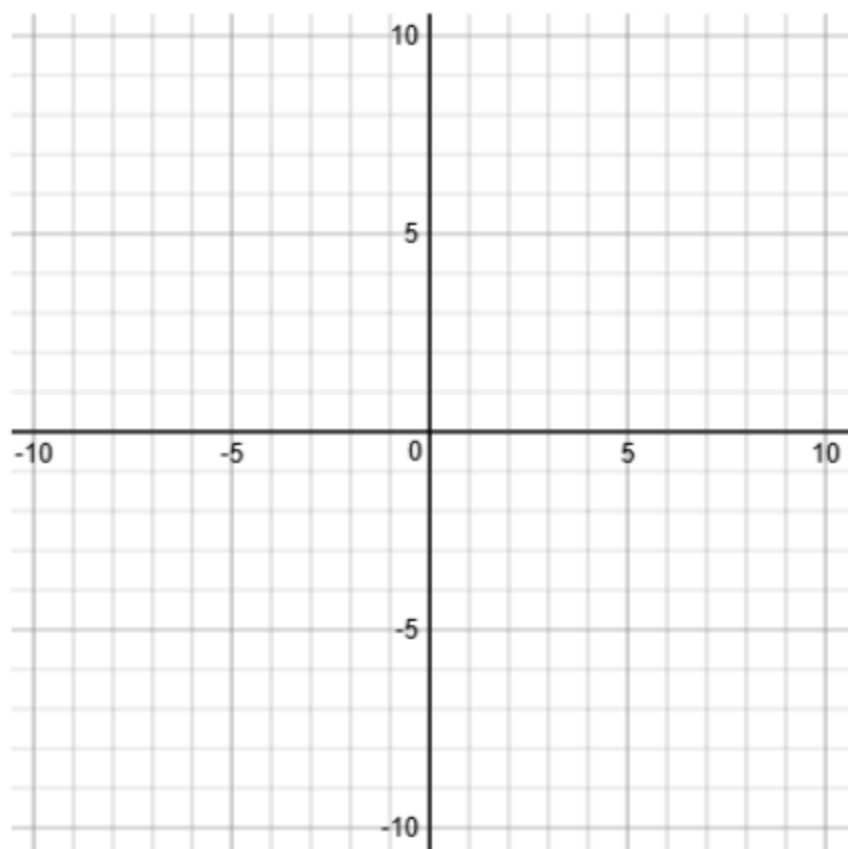


$$5) f(x) = \frac{3}{x^2 - 4}$$

a) Graph $f(x) = \frac{3}{x^2 - 4}$.

Vertical Asymptotes are:

$$x = \underline{\quad ? \quad} \text{ and } \underline{\quad ? \quad}$$



b) $\lim_{x \rightarrow -2^-} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches -2 from the left

$$\lim_{x \rightarrow -2^+} f(x) = \underline{\quad ? \quad}$$

limit of $f(x)$ as x approaches -2 from the right

$$\lim_{x \rightarrow -2} f(x) = \underline{\quad ? \quad}$$

limit of $f(x)$ as x approaches -2

c) $\lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 2 from the left

$\lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 2 from the right

$\lim_{x \rightarrow 2} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

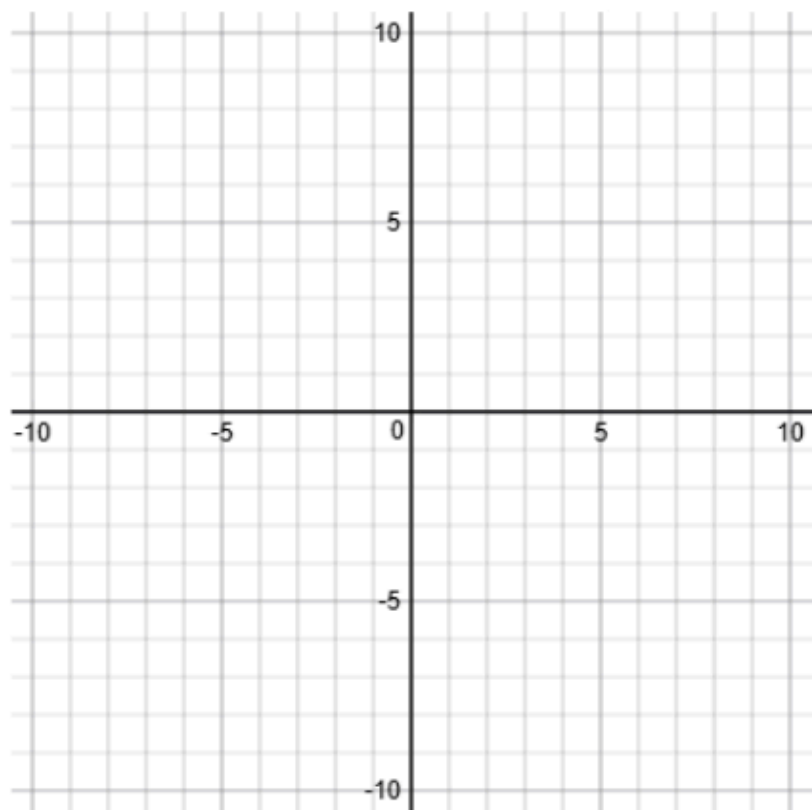
limit of $f(x)$ as x approaches 2

6) $f(x) = \frac{x}{x^2 - 9}$

a) Graph $f(x) = \frac{x}{x^2 - 9}$.

Vertical Asymptotes are:

$x = \underline{\quad ? \quad}$ and $\underline{\quad ? \quad}$



b) $\lim_{x \rightarrow -3^-} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches -3 from the left

$\lim_{x \rightarrow -3^+} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches -3 from the right

$\lim_{x \rightarrow -3} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches -3

c) $\lim_{x \rightarrow 3^-} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 3 from the left

$\lim_{x \rightarrow 3^+} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches 3 from the right

$\lim_{x \rightarrow 3} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

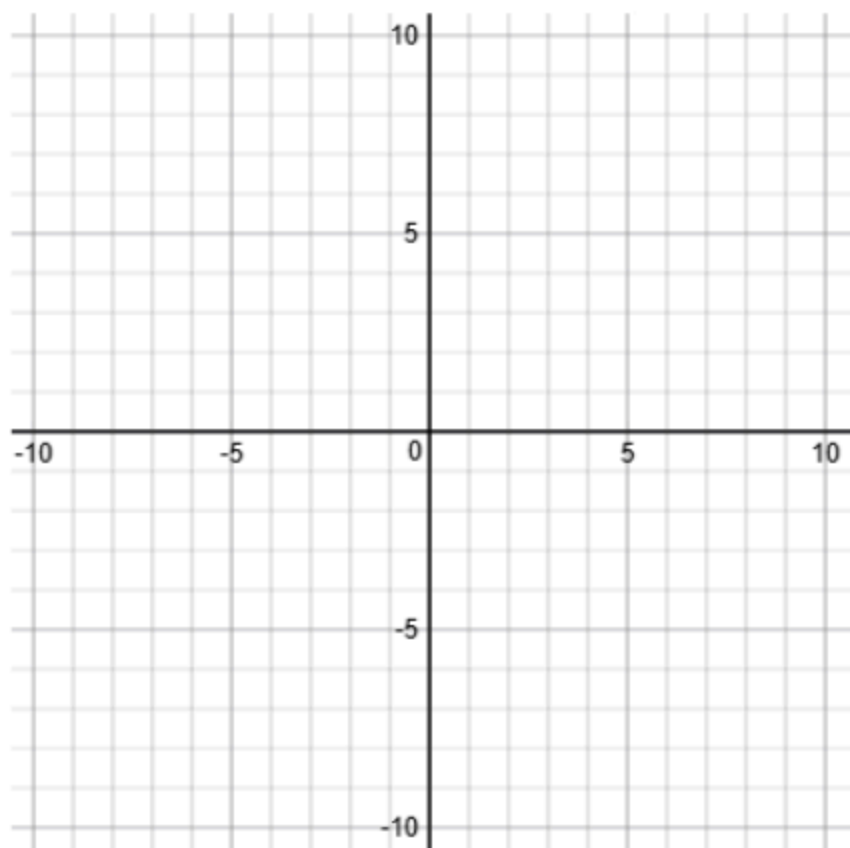
limit of $f(x)$ as x approaches 3

$$7) f(x) = \frac{x^2}{x^2 + 5x - 6}$$

a) Graph $f(x) = \frac{x^2}{x^2 + 5x - 6}$

Vertical Asymptotes are:

$x = \underline{\quad ? \quad}$ and $\underline{\quad ? \quad}$



b) $\lim_{x \rightarrow -6^-} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches -6 from the left

$\lim_{x \rightarrow -6^+} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches -6 from the right

$\lim_{x \rightarrow -6} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches -6

c) $\lim_{x \rightarrow 1^-} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches 1 from the left

$\lim_{x \rightarrow 1^+} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches 1 from the right

$\lim_{x \rightarrow 1} f(x) = \underline{\quad ? \quad}$

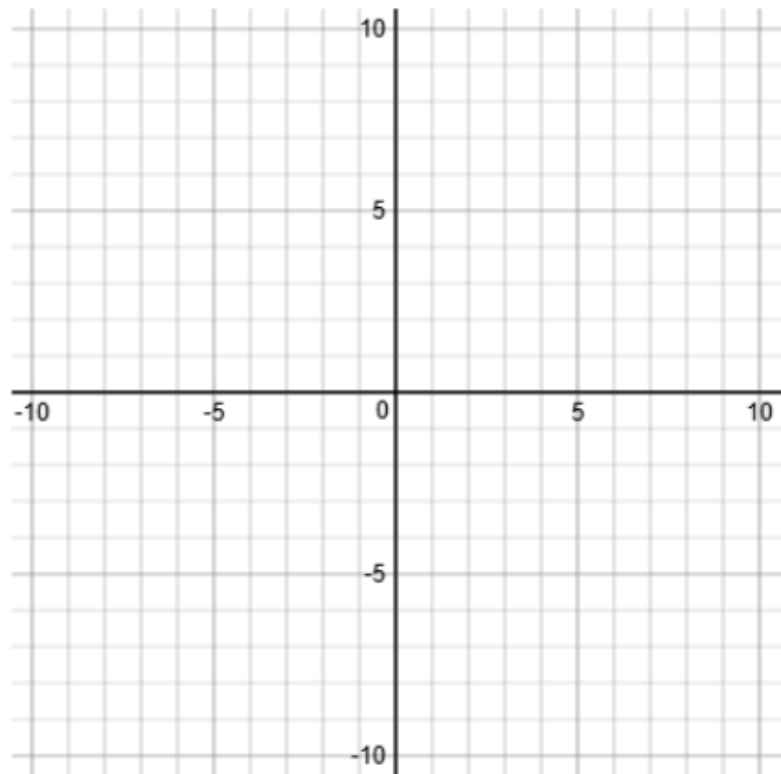
limit of $f(x)$ as x approaches 1

$$8) f(x) = \frac{4x}{x^2 - 3x - 4}$$

$$a) \text{ Graph } f(x) = \frac{4x}{x^2 - 3x - 4}.$$

Vertical Asymptotes are:

$x = \underline{\quad ? \quad}$ and $\underline{\quad ? \quad}$



$$b) \lim_{x \rightarrow 4^-} f(x) = \underline{\quad ? \quad}$$

limit of $f(x)$ as x approaches 4 from the left

$$\lim_{x \rightarrow 4^+} f(x) = \underline{\quad ? \quad}$$

limit of $f(x)$ as x approaches 4 from the right

$$\lim_{x \rightarrow 4} f(x) = \underline{\quad ? \quad}$$

limit of $f(x)$ as x approaches 4

c) $\lim_{x \rightarrow -1^-} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches -1 from the left

$\lim_{x \rightarrow -1^+} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches -1 from the right

$\lim_{x \rightarrow -1} f(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

limit of $f(x)$ as x approaches -1

